MOLAR RATIOS

A balanced equation shows us how many moles of each substance are used or produced in a chemical reaction. Complete the questions in a similar way to the example which has been done for you.

EXAMPLE	C_3H_8	+	$5O_2$	→	$3CO_2$	+	$4H_2O$
what it means	1 mol of C_3H_8		5 mol of O_2		3 mol of CO_2		4 mol of H_2O
a)	2 mol		10 mol		6 mol		8 mol
b)	10 mol		50 mol		30 mol		40 mol
c)	0.5 mol		2.5 mol		1.5 mol		2.0 mol

1	$2Ca$	+	O_2	→	$2CaO$
what it means	**2 mol of Ca**		**1 mol of O_2**		**2 mol of CaO**
a)	4 mol		**2 mol**		**4 mol**
b)	**10 mol**		5 mol		**10 mol**
c)	0.10 mol		**0.05 mol**		**0.10 mol**

2	$2Fe$	+	$3Cl_2$	→	$2FeCl_3$
what it means	**2 mol of Fe**		**3 mol of Cl_2**		**2 mol of $FeCl_3$**
a)	10 mol		**15 mol**		**10 mol**
b)	**6 mol**		9 mol		**6 mol**
c)	**0.40 mol**		0.60 mol		**0.40 mol**

3	$TiCl_4$	+	$4Na$	→	Ti	+	$4NaCl$
what it means	**1 mol of $TiCl_4$**		**4 mol of Na**		**1 mol of Ti**		**4 mol of NaCl**
a)	3 mol		**12 mol**		**3 mol**		**12 mol**
b)	**2.5 mol**		10 mol		**2.5 mol**		**10 mol**
c)	0.020 mol		**0.080 mol**		**0.020 mol**		**0.080 mol**

4	$2Al_2O_3$	→	$4Al$	+	$3O_2$
what it means	**2 mol of Al_2O_3**		**4 mol of Al**		**3 mol of O_2**
a)	5 mol		**10 mol**		**7.5 mol**
b)	**0.050 mol**		0.100 mol		**0.075 mol**
c)	**40 mol**		**80 mol**		60 mol

5	C_2H_5OH	+	$3O_2$	→	$2CO_2$	+	$3H_2O$
what it means	**1 mol of C_2H_5OH**		**3 mol of O_2**		**2 mol of CO_2**		**3 mol of H_2O**
a)	**4 mol**		12 mol		**8 mol**		**12 mol**
b)	0.25 mol		**0.75 mol**		**0.50 mol**		**0.75 mol**
c)	**0.05 mol**		**0.15 mol**		0.10 mol		**0.15 mol**